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2027 CYCLE

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CUET UG

Biology

Human Health and Disease

Official Syllabus Unit VIII · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VIII)	Skills the questions test
Health and disease — introduction	Introduction about health;	Telling health from disease, and linking well-being to genes, germs and lifestyle.
Common diseases in humans	Common diseases in humans (typhoid, pneumonia, common cold, malaria, amoebiasis, ascariasis, filariasis and ringworms);	Matching each disease to its pathogen, vector and test; reading symptom lists.
Immunity — innate and acquired	Immunity – innate and acquired immunity,	Classifying defences as inborn or learned, and spotting the four barrier types.
Active, passive, vaccination	active and passive immunity, vaccination and immunisation,	Contrasting active with passive immunity; explaining how a vaccine builds memory.
Allergies and auto-immunity	allergies, auto-immunity and immune system in human	Explaining the IgE–histamine story, autoimmunity, and where immune cells mature.
AIDS	AIDS;	Tracing HIV's attack plan, its transmission rules and the tests that confirm it.
Cancer	Cancer;	Separating benign from malignant growths; naming causes, detection and treatment.
Drugs and alcohol abuse	Drugs and alcohol abuse - adolescence, addiction, effects of drug abuse, prevention and control	Grouping drugs by family and source plant; stating prevention and warning signs.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VIII connects biology to daily life: pathogens, the immune system that fights them, and the habits that keep both in check.

Health, Disease and Pathogens

It is monsoon. The person beside you on the bus sneezes twice without covering the nose, and by Friday three classmates have a fever. Health and disease are not a far-away laboratory story — they are daily life, and this unit is about exactly that: the organisms that attack the body, and the defence system that fights back.

- **Health, the WHO way:** the World Health Organization defines health as a state of **total well-being** — physical, mental and social — not simply "no illness".
- **Disease, in plain words:** a **disease** is any condition in which the body or the mind stops working normally. The word literally means "not at ease".
- **Pathogens, the attackers:** a **pathogen** is a disease-causing organism — a bacterium, virus, fungus, protozoan (single-celled organism) or worm — that enters the body and multiplies in it.
- **Where health slips:** health problems come from three broad directions — **genetic faults** (inborn, e.g. haemophilia), **infections** (pathogens), and **lifestyle** (junk food, no exercise, alcohol and drugs).
- **Health is a habit:** balanced diet, regular exercise, safe water and hygiene protect you as much as any medicine, because most pathogens enter by mouth, nose or skin.

How diseases travel from one body to another

Route	How it works	Examples
Through the air	The pathogen rides in tiny droplets sprayed out when an infected person coughs, sneezes or talks, and you breathe them in.	Common cold, pneumonia, tuberculosis
Through water and food	The pathogen is swallowed along with contaminated water, unwashed vegetables or food handled by dirty hands.	Typhoid, amoebiasis, ascariasis
Through a vector	A **vector** is an animal — most often an insect — that carries the pathogen from one person to the next without itself falling ill.	Malaria — *Anopheles* mosquito; filariasis — *Culex* mosquito
Through body fluids	The pathogen passes in blood, semen, breast milk or shared needles — direct contact with infected fluids.	AIDS, hepatitis B and C

- **The one-line summary:** pathogens need a ride — dirty water, a sneeze, a mosquito bite or a shared needle — and cutting off that ride cuts off the disease.
- **Watch the wording:** questions mix up these routes on purpose; a water-borne disease is not the same as a vector-borne one.

The body is not defenceless

- **Immunity, the idea: immunity** is the body's overall ability to fight disease-causing organisms and everything else it calls "not self".
- **Two big arms:** some defences are ready **from birth** (innate — Section 3) and others are **learned against each specific enemy** (acquired — Section 4).
- **Why it matters for exams:** nearly half of this unit's syllabus (immunity, allergy, AIDS, cancer) is really about this one system, so learn its logic once and reuse it everywhere.

EVERYDAY EXAMPLE — THE CITY DEFENCE PLAN

Think of your body as a walled city. The **walls and gates** (skin, mucus) stop most invaders at the entrance. The **guards on patrol** (white blood cells) capture anyone who slips in. And the **wanted-poster register** (acquired immunity) remembers every criminal's face, so a second visit ends faster. All three layers appear again in the next sections — keep the city picture.

The Eight Common Diseases

One cold monsoon week can hand you typhoid from water, a cold from a sneeze and malaria from a single mosquito. The syllabus names eight diseases, and for each one three things matter most: the **pathogen**, the **way it spreads**, and **one signature fact** (a test, a symptom or a vaccine). The quick-map below holds all three for every disease — revise from it.

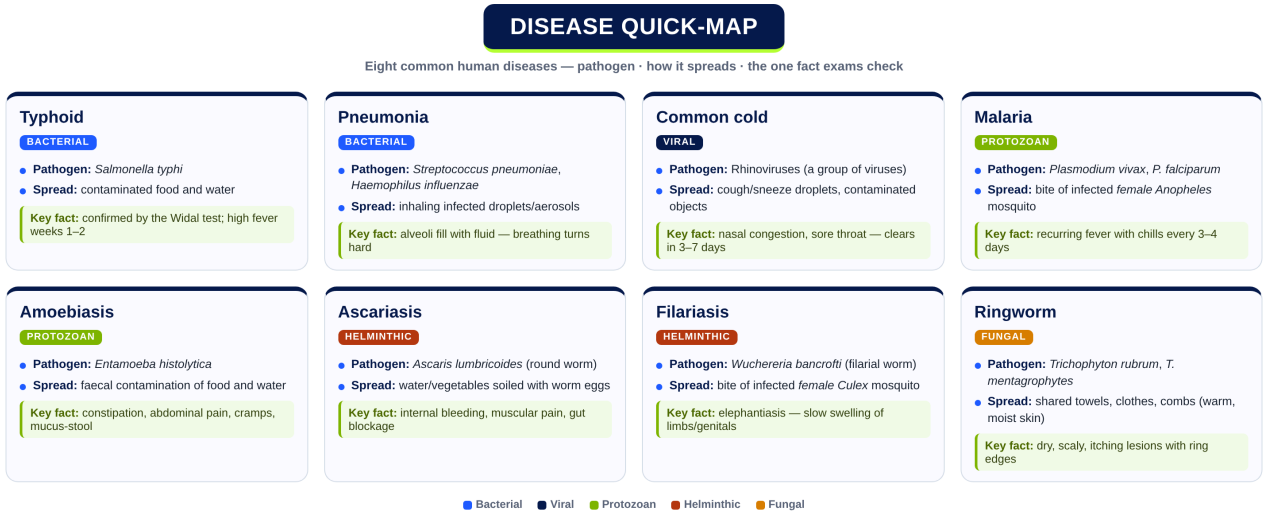


Figure 1: Disease quick-map — the eight common human diseases, each with pathogen, route of spread and the one fact that identifies it.

The full pathogen-to-prevention table

Disease	Pathogen	How it enters	Main symptoms	Prevention
Typhoid	*Salmonella typhi* (bacterium)	Contaminated food and water	Sustained high fever (39–40°C), weakness, stomach pain, constipation	Safe water, food hygiene, typhoid vaccine
Pneumonia	*Streptococcus pneumoniae*, *Haemophilus influenzae* (bacteria)	Inhaling infected droplets	Fever, chills, cough, headache; lips turn blue in severe cases	Avoid close contact with patients, protect from cold
Common cold	Rhinoviruses (viruses)	Droplets from coughs and sneezes	Nasal congestion, sore throat, cough, hoarseness, tiredness	Hand washing, avoiding shared towels and glasses
Malaria	*Plasmodium vivax*, *P. falciparum* (protozoan)	Bite of infected *female Anopheles* mosquito	Recurring chill and high fever every 3–4 days, sweating	Vector control, nets and repellents, no stagnant water
Amoebiasis	*Entamoeba histolytica* (protozoan)	Faecal contamination of food and water	Constipation, abdominal pain and cramps, mucus and blood in stools	Safe drinking water, washed vegetables, fly control
Ascariasis	*Ascaris lumbricoides* (helminth worm)	Eggs swallowed in contaminated water or vegetables	Internal bleeding, muscular pain, fever, anaemia, gut blockage	Washing produce, deworming, sanitation
Filariasis	*Wuchereria bancrofti* (helminth worm)	Bite of infected *female Culex* mosquito	Chronic swelling of limbs — elephantiasis; genital organs may be affected	Mosquito control, nets, treating patients early
Ringworm	*Trichophyton rubrum*, *T. mentagrophytes* (fungi)	Shared towels, clothes, combs; skin contact	Dry, scaly, intensely itchy lesions on skin, nails and scalp	Personal hygiene, no sharing of towels and clothes

One-line cards: the signature fact of each disease

- **Typhoid — the fever with a test:** confirmed by the **Widal test**, a blood test that spots typhoid antibodies; severe untreated cases can perforate the intestine wall.

- **Pneumonia — flooded air sacs:** the bacteria infect the **alveoli** — the tiny balloon-like air sacs in the lungs — which fill with fluid, so oxygen exchange struggles.
- **Common cold — fast and self-clearing:** a group of **rhinoviruses** attacks the nose lining; it simply clears in **3–7 days**, needing rest and fluids, not antibiotics.
- **Malaria — the 3–4 day clock:** the parasite multiplies inside red blood cells; when they burst, the toxin **haemozoin** is released and brings the chill-fever cycle. Quinine, from the bark of the **Cinchona** tree, was the first anti-malarial drug.
- **Amoebiasis — the gut invader:** the protozoan feeds on the intestine lining; the **housefly** often acts as a mechanical carrier from faeces to food.
- **Ascariasis — the intestinal worm:** swallowed eggs hatch into adult round worms inside the intestine; a heavy load can physically **block the gut**.
- **Filariasis — elephantiasis:** the worms live in **lymph vessels** (the channels that drain tissue fluid), so fluid builds up and limbs swell slowly over years.
- **Ringworm — the itchy ring:** heat and moisture help the fungus grow, which is why sweaty clothes, skin folds and shared gym towels spread it.

Match-the-pair drill: disease against pathogen

Disease	Correct partner (pathogen + type)
Typhoid	*Salmonella typhi* — bacterium
Pneumonia	*Streptococcus pneumoniae* — bacterium
Common cold	Rhinovirus — virus
Malaria	*Plasmodium* — protozoan
Amoebiasis	*Entamoeba histolytica* — protozoan
Ascariasis	*Ascaris lumbricoides* — helminth
Filariasis	*Wuchereria bancrofti* — helminth
Ringworm	*Trichophyton* — fungus

Immunity I — Innate: The Defence You Are Born With

A newborn has never met a single germ, yet survives a world full of them. The reason is **innate immunity** — the defence system you are born with, which attacks **any** pathogen on sight, without caring which species it is. Because it is non-specific and has no memory, it is the body's first, fastest response layer.

- **Non-specific means:** the same defence fights **Salmonella**, rhinovirus and a fungus alike — no name-taking, no special weapons for each enemy.
- **From birth:** it does not need a prior meeting with the pathogen; it works on day one of life.
- **No memory:** repeat the same attack and the response is **not** faster or stronger the second time — this single line decides many exam answers.

The four barrier types

Barrier type	How it blocks pathogens	Examples from the body
Physical	Sheets of cells that mechanically block entry — pathogens simply cannot cross the wall.	Skin covering the body; mucus coating the airway, gut and urogenital lining
Physiological	Body fluids with conditions or chemicals that kill or slow pathogens.	Acid (HCl) in the stomach; **lysozyme** (an enzyme that digests bacterial walls) in saliva and tears
Cellular	Special white blood cells that swallow and digest invaders — a process called **phagocytosis**.	**PMNL-neutrophils** (white blood cells with lobed nuclei) and **macrophages** (tissue scavenger cells)
Cytokine	Infected cells send out warning proteins called **cytokines** that protect neighbouring cells.	**Interferons** released by virus-infected cells make nearby cells virus-resistant

- **The label trick:** match the noun to the type — a **wall** is physical, a **chemical** is physiological, a **cell** is cellular, a **message** between cells is a cytokine barrier.
- **Interferons, one line deeper:** they "interfere" with virus spread — the infected cell tells its neighbours how to lock their doors before the virus arrives.

EVERYDAY EXAMPLE — THE MEDIEVAL CASTLE

A castle under siege maps one-to-one onto innate immunity: the **stone walls** are the skin, the **boiling oil poured on climbers** is the stomach's acid and other killing fluids, the **guards who throw attackers into the moat** are the phagocytes (neutrophils and macrophages), and the **watchtower drum sent to warn the next village** is an interferon. Notice the castle never learns the attacker's name — that is exactly why innate immunity has no memory.

- **Where it ends:** innate immunity buys time but cannot finish a clever enemy — for that, the body needs targeted, remembered defence: acquired immunity.

Immunity II — Acquired, Active and Passive, Vaccination

The innate system treats every enemy the same. Sooner or later, though, the body needs a sniper rather than a wall — something aimed at **this particular** pathogen. That is **acquired immunity**: it develops after meeting a pathogen (or a vaccine), it is **pathogen-specific**, and it carries **memory**, so the second meeting ends much faster than the first.

- **The specialised cells:** two families of **lymphocytes** (a type of white blood cell) run this system — **B-lymphocytes** and **T-lymphocytes**.

- **The weapons:** B cells produce **antibodies** — Y-shaped proteins that stick to one specific antigen (**antigen** = any foreign molecule that the immune system recognises as "not self").
- **Millions of targets:** the body can make antibodies against **millions of different antigen types**, which is why one system covers a lifetime of new enemies.

The antibody: a Y with five flavours

- **Shape first:** every antibody is a **Y-shaped protein** — the two arms of the Y grip the antigen; the stem does the rest of the work.
- **Five classes, one role each:** IgA, IgM, IgG, IgE and IgD — the letters matter, and each class has its own job:

Class	Its job in one line	Where you meet it
IgG	Crosses the placenta to protect the baby before birth.	The most abundant antibody in blood
IgA	Guards wet surfaces and passes to the new-born in milk.	**Colostrum** — the yellowish first milk, days 2–3
IgM	The **first responder** — first antibody made in a fresh infection.	Early blood test of a new infection
IgE	Drives **allergic reactions** — see Section 5.	Allergy to pollen or dust
IgD	Sits on B cells as their antigen sensor.	Activates B-lymphocytes

Humoral vs cell-mediated: two arms of acquired immunity

Arm	Who does the work	What it handles
Humoral	**B-lymphocytes** make antibodies that float in blood and lymph (body fluids — hence "humoral").	Free bacteria, toxins, viruses outside cells
Cell-mediated	**T-lymphocytes** act directly — no antibodies. Helper T cells also boost B cells.	Virus-infected cells, fungi, and **graft rejection** — the body rejecting a transplanted organ

- **The transplant line:** a donated kidney is rejected by **cell-mediated** immunity — T cells recognise the graft's antigens as foreign. This is a classic match-the-pair fact.

Primary vs secondary response: the reason vaccines work

- **Primary response:** the **first** meeting with an antigen — antibodies rise slowly and only to a modest level, while the pathogen does its damage.
- **Memory cells:** some B and T cells stay behind after the fight, standing by specifically for that antigen.
- **Secondary response:** the **second** meeting — memory cells wake instantly, so antibodies rise **faster** and **far higher**, often killing the pathogen before you feel ill.

- **The exam line:** a vaccine works because it manufactures this **memory without the disease**.

WORKED EXAMPLE — READING A RESPONSE CURVE IN WORDS

A student receives an anti-tetanus vaccine in January and a booster in July.

- **January (primary):** the body meets the tetanus antigen for the first time — antibody level climbs slowly over 1–2 weeks, peaks low, and fades.
- **July (secondary):** memory B cells recognise the same antigen at once — antibody level rises **within days** to a much higher, longer-lasting peak.
- **Conclusion:** the booster is faster and stronger — not because the vaccine changed, but because the **defence changed**: it now has memory.

Active vs passive: who made the antibodies?

Point	Active immunity	Passive immunity
Who makes the antibodies	Your **own body**, after meeting the antigen.	Ready-made antibodies **given** from outside.
How it is gained	Natural infection, or a **vaccination** (dead/weakened antigen).	Colostrum IgA for the baby; **antivenom** after a snakebite.
Memory	Present — protection lasts **years**.	Absent — protection is **fast but short-lived**.
One-line check	Slow to arrive, long to stay.	Instant to arrive, quick to fade.

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- **Vaccination and immunisation, defined: vaccination** is giving a weakened or dead antigen so the body builds memory; **immunisation** is the whole process — the body ends up protected against that disease.
- **Modern twist:** some vaccines (e.g. hepatitis B) are made by **recombinant DNA technology** — the antigen protein is grown in yeast cells, so no live pathogen is needed.

Allergies, Autoimmunity and the Immune System

The immune system's whole job is telling **self** from **non-self**. When it over-reacts to something harmless, you get an **allergy**. When it mistakes self for enemy, you get **autoimmunity**. And when you ask "where do all these soldiers come from?", the answer is the **lymphoid organs** — the barracks and training grounds of immunity.

Allergy: a defence that over-fires

- **Definition:** an **allergy** is an **exaggerated** immune response to a normally harmless substance in the environment.

- **Allergens, the triggers: allergen** is the name for such a substance — common ones are **pollen grains** and **dust mites**; some people react to animal dander (flakes of skin and fur).
- **The chemistry chain:** the first exposure makes **IgE antibodies** against the allergen; on the next exposure, these IgE antibodies make body cells release **histamine and serotonin** — the chemicals that actually cause the symptoms.
- **The symptoms:** sneezing, watery eyes, running nose, difficulty in breathing — all in minutes, which is why allergy feels so sudden.
- **The medicine logic: antihistamines** (plus adrenaline and steroids in strong cases) reduce the symptoms by blocking the released chemicals — this is why an anti-allergy tablet is called an antihistamine.

Autoimmunity: the army attacks its own city

- **The normal rule:** higher vertebrates have the ability to tell **self-cells** from foreign cells and leave the self alone.
- **The breakdown:** sometimes, due to genetic and other unknown reasons, the body **attacks its own cells** — this is **autoimmunity**.
- **The named example: rheumatoid arthritis** — the immune system attacks the joints, causing painful inflammation — is the standard auto-immune disease of this chapter.
- **The exam line:** autoimmunity is not weak immunity; it is **misdirected** immunity — strong but aimed wrongly.

Lymphoid organs: where immune cells are made and used

Organ	Class	Its role in one line
Bone marrow	**Primary**	Birthplace — all blood cells, including lymphocytes, are produced here.
Thymus	**Primary**	Training school — T-lymphocytes **mature** here; it shrinks with age.
Spleen	**Secondary**	Blood filter — traps blood-borne germs; the **graveyard of RBCs** and a large reservoir of lymphocytes.
Lymph nodes	**Secondary**	Checkpoints along lymph vessels — trap germs and are sites where lymphocytes meet antigen.
Tonsils, Peyer's patches	**Secondary**	Guard posts at the entrances — tonsils in the throat, Peyer's patches in the small-intestine wall.

- **Primary vs secondary, in one line: primary** organs **make and mature** lymphocytes; **secondary** organs are the **battlefields** where mature lymphocytes meet antigens and multiply.

- **The spleen double-role:** it both filters worn-out red blood cells (hence "graveyard of RBCs") and stores lymphocytes ready for blood-borne invaders.

AIDS

AIDS is not one disease — it is the **stage** where the immune system itself has been disabled, so ordinary infections turn dangerous. Understanding one virus explains the whole story.

The virus and its attack plan

- **The culprit:** AIDS (Acquired Immuno Deficiency Syndrome) is caused by **HIV** — the Human Immuno-deficiency Virus.
- **Retrovirus, explained:** HIV is a **retrovirus** — its genetic material is **RNA**, and an enzyme called **reverse transcriptase** copies that RNA **into DNA**, which then slips into the host cell's own genome.
- **The target:** HIV selectively infects and destroys **helper T-lymphocytes** — the very cells that coordinate the rest of the immune system.
- **The result:** helper T cell count falls, immunity collapses, and the patient cannot fight **opportunistic infections** — mild germs (certain bacteria, fungi, parasites) that a healthy body would shrug off.

How HIV travels — and how it does not

HIV DOES spread by	HIV does NOT spread by
Sexual contact with an infected person	Handshakes, hugs or sitting beside the person
Transfusion of infected blood or blood products	Sharing meals, water or utensils
Sharing infected needles (e.g. among drug users)	Mosquito or insect bites
Infected mother to the baby through the placenta	Sweat, tears, coughs and sneezes

- **Why the left column works:** every real route carries **body fluids** — the virus lives in blood, semen and breast milk, and dies quickly outside them.
- **Why the right column fails to spread:** those contacts exchange no body fluids, which is exactly what the common myths get wrong.

Tests, timeline and prevention

- **Screening first: ELISA** (Enzyme-Linked Immuno-Sorbent Assay) is the widely used **screening** test — cheap and quick for large numbers.
- **Confirmation next:** a positive ELISA is confirmed by the **Western blot** test; **PCR** (which detects viral genetic material) is the third tool.
- **The timeline:** the gap between infection and AIDS symptoms varies from a few months to many years — **often about 5–10 years** — because the virus works slowly; the person may look and feel normal while carrying it.

- **Prevention, the official list:** safe sexual practices, use of condoms, blood tested before transfusion, disposable needles and syringes, and avoiding drug abuse.
- **The programmes:** in India, **NACO** (National AIDS Control Organisation) and globally the **WHO** run awareness, testing and blood-safety programmes.

EVERYDAY EXAMPLE — THE COMMAND CENTRE SABOTAGE

HIV does not fight the army — it **kills the generals**. Helper T cells are the command centre that orders B cells and killer T cells into action. With the generals gone, even a mild infection (a foot soldier's skirmish) can overrun the city — that is what "opportunistic infection" means: the germ seizes the **opportunity** left by a defenceless body.

Cancer

Your cells are polite: when they touch a neighbour, they stop dividing. This property — **contact inhibition** — keeps every tissue a tidy single layer. Cancer is what happens when cells lose their manners.

How a normal cell turns cancerous

- **The rule: normal cells show contact inhibition** — they stop dividing as soon as they touch neighbouring cells.
- **The break: cancer cells lose contact inhibition**, so they keep dividing without pause, piling on each other in a growing mass.
- **The trigger:** under certain conditions, normal genes called **proto-oncogenes** get activated into **oncogenes** (cancer-causing versions), pushing the cell into uncontrolled division.

Benign vs malignant — and the word metastasis

Point	Benign tumour	Malignant tumour (cancer)
Growth speed	Slow and contained.	Rapid and aggressive.
Stays put?	Yes — confined to its original location.	No — invades neighbouring tissues.
Spreads to new sites?	**Never** — this is the defining difference.	**Yes** — this spread is called **metastasis**.
Danger level	Lower; often removable.	High; the spread is what kills.

- **Metastasis, defined:** malignant cells travel through **blood and lymph** to new organs and start fresh tumours there — this travel is **metastasis**.

Causes: the carcinogen families

- **Physical agents:** radiation — X-rays, gamma rays, UV rays — damages the genes that control division.

- **Chemical agents: chemical carcinogens**, with **tobacco smoke** as the major example (its link to lung cancer is the classic case).
- **Biological agents: oncogenic viruses** — viruses carrying genes that can turn a cell cancerous.

Detection and treatment

Detection method	What it does
Biopsy + histopathology	A piece of the suspected tissue is examined under the microscope — the **confirming** test.
Radiography / CT / MRI	X-ray imaging and scan techniques locate tumours inside the body non-invasively.
Blood and bone-marrow tests	Reveal blood-cell cancers such as leukaemias.

Treatment	How it attacks the tumour
Surgery	The tumour is physically removed — works when it is still localised.
Radiation therapy	Focused rays kill the rapidly dividing cells in the tumour zone.
Chemotherapy	Anti-cancer drugs circulate in blood and kill dividing cells (side effects: hair loss, anaemia).
Immunotherapy	**Biological response modifiers** such as **α-interferon** activate the patient's own immune system against the tumour.

- **The mole rule:** a mole that changes shape, size or colour, or starts bleeding, needs a doctor's look — early detection is the single biggest survival factor.

Drugs and Alcohol Abuse

Between 12 and 18 you are neither child nor adult — a bridge years long, with exams, new freedoms and a strong wish to fit in. **Adolescence** (those bridge years) is exactly when curiosity and peer pressure push many towards tobacco, alcohol and drugs. This section names the substances, the plants they come from, and the exits available.

Addiction and tolerance: how the trap closes

- **Addiction, defined: addiction** is physical and psychological dependence — the person keeps using the substance although it is damaging health, studies and family.
- **Tolerance, defined:** with repeated use, the body **tolerates** the old dose, so the person needs a **higher and higher dose** to feel the same effect — a self-reinforcing spiral.
- **Withdrawal:** stopping suddenly brings anxiety, shakiness, nausea and sweating — the body's protest, and the reason quitting alone is so hard.
- **Why it starts:** curiosity, the wish to appear grown-up, stress and escape, and above all **peer pressure** — the push to do what the group does.

The three drug families

Family	Example and source plant	Action on the body
Opioids	**Heroin** ("smack") — extracted from the latex of the poppy plant *Papaver somniferum*.	**Depressant** — binds **opioid receptors** in the central nervous system and gut, slowing body functions
Stimulants	**Cocaine** ("coke" or "crack") — from the coca plant *Erythroxylum coca*.	**Stimulates** the CNS — euphoria and extra energy; interferes with dopamine transport; high doses cause hallucinations
Cannabinoids	Marijuana, charas, ganja — from the hemp plant *Cannabis sativa*.	**Hallucinogen** — interacts with **cannabinoid receptors** in the brain, distorting perception; strains the heart

- **The one-line key:** opioids **slow**, stimulants **rev**, cannabinoids **distort** — three verbs that sort every question in this area.
- **Tobacco, the quiet entry: nicotine** stimulates the adrenal glands, raising blood pressure and heart rate, and is linked to lung cancer, bronchitis, emphysema and coronary heart disease.
- **Alcohol, the liver tax:** heavy drinking destroys liver cells (cirrhosis) and damages the nerves and the family that lives with the drinker.

The cost beyond the body

- **On the person:** falling marks, lost appetite and sleep, weak immunity, accidents and violence.
- **On the family:** money drains away, trust breaks, and the household lives around one person's craving.
- **On society:** drug-dependent youth mean lost careers, crime and pressure on health services — which is why prevention is a national programme, not a personal choice.

Prevention and the way out

- **Avoid undue peer pressure:** it is fine to say no when the group pushes — the confidence to refuse is itself a skill.
- **Channel energy:** sports, music, hobbies and studies fill the same hours that risk-taking tries to fill.
- **Education and counselling:** knowing the exact damage (this section) plus talking to counsellors catches problems early.
- **De-addiction centres:** doctors and trained staff manage withdrawal safely and support recovery — seeking help is treatment, not shame.
- **Warning signs to act on:** sudden fall in grades, secrecy about friends and money, red eyes, changed sleep and appetite, unexplained mood swings.

Key Facts & Terms Cheatsheet

KEY FACTS & TERMS CHEATSHEET — HUMAN HEALTH AND DISEASE

Term / fact	Meaning in one line	Example / number
Health (WHO)	Total well-being — physical, mental and social.	Not merely "no disease"
Typhoid	*Salmonella typhi* via contaminated food and water.	**Widal test**; fever 39–40°C
Pneumonia	*Streptococcus pneumoniae* or *H. influenzae* via droplets.	Alveoli fill with fluid
Common cold	Rhinoviruses via droplets and shared objects.	Self-clears in 3–7 days
Malaria	*Plasmodium vivax* / *P. falciparum* protozoan.	**Female Anopheles**; chill-fever every 3–4 days
Amoebiasis	*Entamoeba histolytica* via faecal contamination.	Mucus and blood in stools
Ascariasis	*Ascaris lumbricoides* eggs in water/vegetables.	Gut blockage, anaemia
Filariasis	*Wuchereria bancrofti* worms in lymph vessels.	**Female Culex**; elephantiasis
Ringworm	*Trichophyton* fungi from shared towels/clothes.	Dry, scaly, itchy lesions
Physical barrier	Blocks entry mechanically.	Skin, mucus lining
Physiological barrier	Killing conditions in body fluids.	Stomach HCl; lysozyme in saliva/tears
Cellular barrier	Cells that swallow pathogens (phagocytosis).	PMNL-neutrophils, macrophages
Cytokine barrier	Warning proteins from infected cells.	Interferons
IgG	Antibody that crosses the placenta.	Protects the foetus
IgA	Guards wet surfaces; passes in first milk.	Colostrum, days 2–3
IgM	First antibody made in a new infection.	The first responder

Term / fact	Meaning in one line	Example / number
IgE	The allergy antibody — triggers histamine.	Pollen and dust allergy
Active immunity	Own body makes antibodies; memory forms.	Vaccination; lasts years
Passive immunity	Ready-made antibodies given; no memory.	Antivenom; colostrum IgA
Allergy chemistry	IgE makes cells release histamine and serotonin.	Antihistamine tablets
Autoimmunity	Immune system attacks self-cells.	Rheumatoid arthritis
Primary lymphoid organs	Sites where lymphocytes are made and mature.	Bone marrow, thymus
Secondary lymphoid organs	Sites where lymphocytes meet antigen.	Spleen, lymph nodes, tonsils, Peyer's patches
HIV attack plan	RNA → DNA by reverse transcriptase; kills helper T cells.	Immunity falls; opportunistic infections
HIV tests	ELISA screens; Western blot confirms; PCR detects genome.	Symptoms after 5–10 years
Contact inhibition	Normal cells stop dividing on touching neighbours.	Cancer cells lose it
Metastasis	Cancer cells travel by blood/lymph to new sites.	Malignant only — never benign
Carcinogen types	Physical, chemical and biological agents.	X-rays; tobacco smoke; oncogenic viruses
Cancer treatments	Four routes against the tumour.	Surgery, radiation, chemotherapy, α-interferon
Opioids	Depressants that bind opioid receptors.	Heroin — *Papaver somniferum*
Stimulants	Excite the central nervous system.	Cocaine — *Erythroxylum coca*
Cannabinoids	Hallucinogens acting on brain receptors.	*Cannabis sativa*

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is actually right	Why / the trap
Malaria is spread by the *Culex* mosquito	*Female Anopheles* carries *Plasmodium*.	*Culex* belongs to **filariasis** — the classic vector swap.
Innate immunity remembers a pathogen	Only **acquired** immunity has memory.	Innate is non-specific and fixed from birth.
Passive immunity protects for years	It is **fast but short-lived**.	No memory cells are made — antivenom fades in weeks.
Colostrum gives the baby active immunity	It is **passive** — ready-made IgA.	The mother's antibodies enter, the baby makes none.
IgE is the antibody that fights viruses	IgE drives **allergy**; histamine causes the symptoms.	Interferons are the antiviral warning proteins.
HIV spreads by handshakes, meals or mosquitoes	Only **body fluids** carry HIV.	Sexual contact, infected blood, needles, mother-to-foetus — nothing else.
A benign tumour metastasises	Only **malignant** tumours metastasise.	Staying put is the **defining** property of benign.
Cannabinoids are stimulants	They are **hallucinogens**.	Cocaine is the stimulant; *Cannabis* distorts perception.

WORKED EXAMPLE — THREE ILL STUDENTS, THREE COMPLETE ANSWERS

Three hostelmates fall ill in the same week. A complete answer names **pathogen + route + signature fact** for each.

- **Student A drank tank water at a fair, then ran a 39–40°C fever for days:** typhoid — *Salmonella typhi* — confirm with the **Widal test**.
- **Student B is back from a forest trip with chills and fever every 3–4 days:** malaria — *Plasmodium* through the **female Anopheles** — the fever recurs as infected red blood cells burst and release haemozoin.
- **Student C shared a hostel towel and now has a dry, scaly, itchy ring:** ringworm — *Trichophyton*, a **fungus** — warm, moist skin helped it grow.

> **The method:** whenever a disease question appears, silently ask all three — **which pathogen, how did it enter, what is the signature?** — and the answer writes itself.

The Whole Chapter on One Page

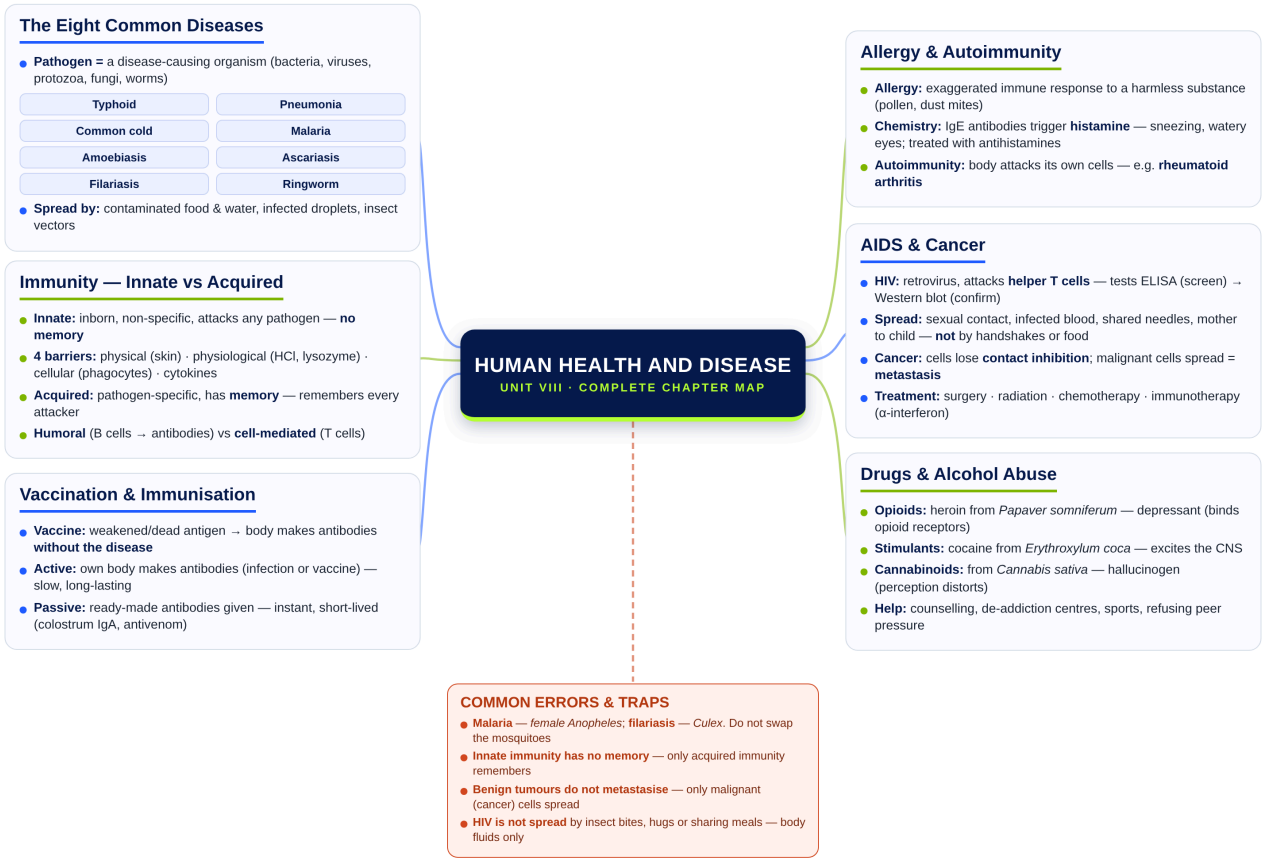


Figure 2: Whole-chapter mindmap — the eight diseases, both arms of immunity, vaccination, allergy and autoimmunity, AIDS and cancer, drugs and alcohol, with the four traps that catch most students.

Practice Questions

Q1. Which disease–pathogen pair is **correctly** matched?

(A) Typhoid — *Salmonella typhi*
(B) Malaria — *Culex* mosquito
(C) Amoebiasis — *Wuchereria bancrofti*
(D) Ringworm — *Plasmodium*

Q2. The malarial parasite reaches the human body through the bite of:

(A) male *Anopheles* mosquito
(B) female *Culex* mosquito
(C) female *Aedes* mosquito
(D) female *Anopheles* mosquito

Q3. Interferons, the proteins released by virus-infected cells, are an example of which innate barrier?

(A) Physical barrier **(B)** Physiological barrier
(C) Cytokine barrier **(D)** Cellular barrier

Q4. Which of the following is an example of **passive** immunity?

(A) IgA antibodies received through colostrum **(B)** Antibodies made after vaccination
(C) Memory B cells formed after an infection **(D)** Skin blocking the entry of bacteria

Q5. HIV disables the immune system mainly by destroying which cells?

(A) Red blood cells **(B)** Helper T-lymphocytes
(C) Blood platelets **(D)** Neutrophils

Q6. A snakebite victim is injected with antivenom — antibodies prepared in another animal's body. The immunity gained is:

(A) active and natural **(B)** active and artificial
(C) passive and artificial **(D)** innate

Q7. Which statement about cancer cells is correct?

(A) They show contact inhibition **(B)** They lose contact inhibition and divide uncontrollably
(C) Every tumour spreads by metastasis **(D)** Benign tumours invade new body sites

Q8. Cocaine belongs to which drug family, and from which plant is it obtained?

(A) Opioid — *Papaver somniferum* **(B)** Cannabinoid — *Cannabis sativa*
(C) Stimulant — *Erythroxylum coca* **(D)** Depressant — *Atropa belladonna*

Q9. The antibody class that crosses the placenta and protects the developing foetus is:

(A) IgG **(B)** IgA
(C) IgM **(D)** IgE

Q10. Which of the following is an **auto-immune** disease?

(A) Typhoid **(B)** AIDS
(C) Filariasis **(D)** Rheumatoid arthritis

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Solutions

Sol. Q1. Typhoid is caused by the bacterium *Salmonella typhi* and confirmed by the Widal test. Malaria is caused by *Plasmodium* carried by **female Anopheles** (not Culex); amoebiasis is caused by *Entamoeba histolytica*; ringworm is caused by fungi such as *Trichophyton*.

Answer: (A)

Sol. Q2. Only the **female Anopheles** bites and transmits *Plasmodium* — male mosquitoes feed on plant juices and do not bite. *Culex* transmits filariasis, and *Aedes* spreads dengue and chikungunya, not malaria.

Answer: (D)

Sol. Q3. Interferons are **cytokines** — warning proteins sent by virus-infected cells to protect their neighbours. Skin is a physical barrier, stomach acid a physiological barrier, and neutrophils/macrophages form the cellular barrier.

Answer: (C)

Sol. Q4. Colostrum, the yellowish first milk, carries **ready-made IgA antibodies** from the mother — the baby's body makes none of them, so this is passive immunity. Vaccination and infection create active immunity; the skin barrier is innate.

Answer: (A)

Sol. Q5. HIV is a retrovirus whose RNA is copied into DNA by reverse transcriptase; the DNA inserts into the genome of **helper T-lymphocytes**, and the infected cells are destroyed. With the coordinators gone, immunity collapses.

Answer: (B)

Sol. Q6. Antivenom gives **ready-made antibodies** produced in another body — protection is instant, carries no memory, and fades quickly, which is the definition of passive immunity. Because it is given as medical treatment rather than by natural transfer (like colostrum), it is **artificial** passive immunity.

Answer: (C)

Sol. Q7. Losing **contact inhibition** — the stop signal that normal cells obey on touching neighbours — is what makes cells cancerous, so they divide without pause. Only **malignant** tumours metastasise; benign tumours stay confined to their original spot.

Answer: (B)

Sol. Q8. Cocaine ("coke" or "crack") is a **CNS stimulant** extracted from the coca plant *Erythroxylum coca* — it speeds up body functions and in high doses causes hallucinations.

Heroin is the opioid from *Papaver somniferum*, and *Cannabis sativa* gives the hallucinogenic cannabinoids.

Answer: (C)

Sol. Q9. IgG is the antibody small enough to cross the placenta, giving the foetus borrowed protection. IgA guards wet surfaces and passes through colostrum; IgM is the first responder; IgE drives allergies.

Answer: (A)

Sol. Q10. An auto-immune disease results when the body attacks **its own** cells — in rheumatoid arthritis the immune system attacks the joints. Typhoid, AIDS and filariasis are all infectious diseases caused by external pathogens, not by self-attack.

Answer: (D)

60-Second Recall Drill

Prompt	Answer
Disease confirmed by the Widal test	Typhoid — *Salmonella typhi*
Mosquito that carries malaria	Female *Anopheles*
Mosquito that carries filariasis	Female *Culex* — causes elephantiasis
Antibody delivered in colostrum	IgA — passive immunity, days 2–3
Antibody that crosses the placenta	IgG
Barrier made of interferons	Cytokine barrier
Phagocyte cells of innate immunity	PMNL-neutrophils and macrophages
HIV test order	ELISA screen → Western blot confirms (+ PCR)
Tumour that metastasises	Malignant only — benign never spreads
Hallucinogen drug and plant	Cannabinoids — *Cannabis sativa*
Depressant from the poppy plant	Heroin — *Papaver somniferum*
Stimulant from the coca plant	Cocaine — *Erythroxylum coca*
“Graveyard of RBCs”	The spleen — filters blood, stores lymphocytes

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